

1 Preliminaries

This chapter develops the necessary preliminaries for what follows, by recalling the notions of principal bundles, classifying spaces and characteristic classes. We assume the reader to be familiar with the concept of a fiber bundle with structure group. A classical reference for this is for example [?]. Moreover, we presuppose some basic familiarity with homotopy and homology theory.

1.1 Principal Bundles and Associated Bundles

Definition 1.1. Let G be a topological group. A G -principal bundle is a fiber bundle with fiber G and structure group G acting by left translations.

The total space of a G -principal bundle $P \xrightarrow{\pi} B$ carries a natural right action of G defined as follows. Around $y \in P$ choose a local trivialisation

$$\phi: U \times G \rightarrow \pi^{-1}(U)$$

and let $\phi^{-1}(y) = (\pi(y), h)$. Then set

$$y \cdot g := \phi(\pi(y), hg).$$

This is well defined since by definition G acts on itself by *left* translations under a change of local trivialisation. A morphism of principal bundles should respect this right action, thus we define:

Definition 1.2. A morphism from a G -principal bundle $P \rightarrow B$ to an H -principal bundle $P' \rightarrow B'$ is a pair (F, f) of continuous maps $F: P \rightarrow P'$ and $f: B \rightarrow B'$ together with a homomorphism $\rho: G \rightarrow H$, such that

$$\begin{array}{ccc} P & \xrightarrow{F} & P' \\ \downarrow & & \downarrow \\ B & \xrightarrow{f} & B' \end{array}$$

commutes and such that F is ρ -equivariant, that is $F(x \cdot g) = F(x) \cdot \rho(g)$ for all $x \in P$ and $g \in G$. If $G = H$ we assume ρ to be the identity and if $B = B'$ we assume f to be the identity.

The condition of equivariance in the above definition is actually quite strong as illustrated by the following proposition.

Proposition 1.3. *Any morphism of G -principal bundles over the same base space is an isomorphism.*

Proof. First suppose we are given a morphism

$$F: B \times G \rightarrow B \times G$$

of trivial bundles. Then by definition F is of the form

$$F(x, g) = (x, \phi_x(g))$$

for a continuous family of equivariant maps $\phi_x: G \rightarrow G$. By equivariance $\phi_x(g) = \phi_x(e)g$ and thus an inverse of F is given by

$$F^{-1}(x, g) := (x, \phi_x(e)^{-1}g).$$

Since every G -principal bundle is locally trivial this shows that we can always find local inverses in the general case and since inverses are unique they fit together to define a global inverse. $\square$

To any fiber bundle with structure group G one can canonically associate a G -principal bundle, obtained by applying a general construction, which can be used to change the fiber of a given bundle. We now recall this construction of changing the fiber. Let $p: E \rightarrow B$ be a fiber bundle with fiber F and structure group G and let F' be another space upon which G acts effectively. Then we can construct a new bundle as follows. Choose an open cover $(U_\alpha)_{\alpha \in A}$ of B together with local trivialisations $\phi_\alpha: U_\alpha \times F \rightarrow p^{-1}(U_\alpha)$. Let $\tau_{\alpha\beta}: U_\alpha \cap U_\beta \rightarrow G$ be the corresponding transition functions defined by

$$(\phi_\beta^{-1} \circ \phi_\alpha)(x, f) = (x, \tau_{\alpha\beta}(x) \cdot f).$$

Define

$$E' := \left(\coprod_{\alpha \in A} U_\alpha \times F' \right) / \sim$$

where the equivalence relation $\sim$ is defined by

$$(b, f', \beta) \sim (b, \tau_{\alpha, \beta}(b) \cdot f', \alpha)$$

and let $p' : E' \rightarrow B$, $[(b, f', \alpha)] \mapsto b$. Then the maps

$$\psi_\alpha : U_\alpha \times F' \rightarrow (p')^{-1}(U_\alpha), (b, f') \mapsto [(b, f', \alpha)].$$

define local trivialisations of p' with the same transition functions $\tau_{\alpha\beta}$, giving $p' : E' \rightarrow B$ the structure of a fiber bundle with fiber F' and structure group G . We say that this new bundle is obtained from $E \rightarrow B$ by *changing the fiber* from F to F' (with the G -action on F' being implicit). Note that we can also obtain $E \rightarrow B$ from $E' \rightarrow B$ by changing the fiber from F' back to F . Thus no information is lost in this process. Considering the special case $F' = G$ with G acting on itself by left translations, we call the bundle obtained from $E \rightarrow B$ by changing the fiber from F to G the *associated principal bundle*. Morally speaking if we want to understand the bundle $E \rightarrow B$ it is sufficient to understand its associated principal bundle, as it contains the same information as we have seen above. The benefit of studying principal bundles is that they are highly symmetric objects, and thus usually easier to understand. For example many constructions simplify for principal bundles, such as the concept of universal bundles as we will see in the next subsection. Another such instance is the construction for changing the fiber from above, which can be carried out more directly in the case of principal bundles and has the benefit of also allowing for non-effective group actions. This is known as the *Borel Construction*.

Lemma 1.4. *Let $p : P \rightarrow B$ be a G -principal bundle and F a space upon which G acts from the left. Then*

$$E := (P \times F) / \sim, \quad (y, f) \sim (y \cdot g, g^{-1}f)$$

together with the map $\pi : E \rightarrow B$ given by $[(y, f)] \mapsto p(y)$ is a fiber bundle with fiber F and structure group G . Moreover, if the action of G on F is effective, this bundle is isomorphic to the bundle obtained from $P \rightarrow B$ by changing the fiber from G to F using the construction outlined above.

Proof. To see that $P \times_G F$ is locally trivial let $\psi : U \times G \rightarrow P$ be a local trivialisation. Then the homeomorphism

$$\psi \times \text{id}_F : U \times G \times F \rightarrow p^{-1}(U) \times F$$

descends to a homeomorphism

$$\tilde{\kappa} : (U \times G \times F) / \sim \rightarrow (p^{-1}(U) \times F) / \sim = \pi^{-1}(U),$$

where the equivalence relation on $U \times G \times F$ is given by $(x, g, f) \sim (x, gh, h^{-1}f)$. Moreover the map

$$U \times G \times F \rightarrow U \times F, (x, g, f) \mapsto (x, gf)$$

descends to a homeomorphism $(U \times G \times F) / \sim \rightarrow U \times F$ with inverse given by $(x, f) \mapsto [(x, e, f)]$.

Under this homeomorphism $\tilde{\kappa}$ corresponds to a homeomorphism $\kappa : U \times F \rightarrow \pi^{-1}(U)$, which is explicitly given by $(x, f) \mapsto [(\psi(x, e), f)]$ and

$$\begin{array}{ccc} U \times F & \xrightarrow{\kappa} & \pi^{-1}(U) \\ & \searrow \text{proj.} & \downarrow \pi \\ & & U \end{array}$$

commutes. So $P \times_G F$ is locally trivial. To see that this construction agrees with the change of fiber construction from above if G acts effectively we have to compute the transition functions of the local trivialisations. Let $U_\alpha, U_\beta \subseteq B$ be two trivialising neighbourhoods for P with local trivialisations ψ_α, ψ_β and transition function $\tau_{\alpha, \beta} : U_\alpha \cap U_\beta \rightarrow G$. Moreover let $\kappa_\alpha, \kappa_\beta$ be the corresponding local trivialisations for $P \times_G F$ as constructed above. Then we compute

$$(\kappa_\alpha^{-1} \circ \kappa_\beta)(x, f) = \kappa_\alpha^{-1}([\psi_\beta(x, e), f]) = (x, \tau_{\alpha, \beta}(x)f).$$

Thus the statement follows from the fact that fiber bundles with structure group are uniquely determined by the transition functions of local trivialisations. $\square$

In view of the above construction one usually denotes the bundle obtained from $P \rightarrow B$ by changing the fiber from G to F as $P \times_G F \rightarrow B$. The Borel construction is natural in the following sense.

Lemma 1.5. *Let $p: P \rightarrow B$ be a G -principal bundle, F a space upon which G acts from the left and $f: B' \rightarrow B$ a continuous map. Then*

$$(f^*P) \times_G F \cong f^*(P \times_G F).$$

Proof. An explicit model for the pullback bundle is

$$f^*P := \{(y, b') \in P \times B' \mid p(y) = f(b')\}.$$

Then

$$(f^*P) \times_G F = \{(y, b', f) \in P \times B' \times F \mid p(y) = f(b')\} / \sim,$$

for $(y, b', f) \sim (y \cdot g, b', g^{-1} \cdot f)$. Moreover

$$\begin{aligned} f^*(P \times_G F) &= \{([y, f], b') \in P \times_G F \times B' \mid p(y) = f(b')\} \\ &\cong \{(y, f, b') \in P \times F \times B' \mid p(y) = f(b')\} / \sim \end{aligned}$$

for $(y, f, b') \sim (y \cdot g, g^{-1} \cdot f, b')$. Clearly these are isomorphic. $\square$

Lets look at some concrete examples for the theory developed above.

Example 1.6. For simplicity we assume all base spaces B to be smooth manifolds in this example.

- i) Let $E \rightarrow B$ be a real rank n vector bundle. Choosing a bundle metric gives $E \rightarrow B$ the structure of a fiber bundle with fiber $\mathbb{R}^n$ and structure group $O(n)$. The associated principal bundle is explicitly given by the *orthonormal frame bundle* $O(E)$ of E , whose fiber over a point $x \in B$ is the space of all ordered orthonormal bases of E_x . Local trivialisations of the canonical map $p: O(E) \rightarrow B$ can be defined as follows. Choose an orthonormal local frame $e_1, \dots, e_n$ of E defined over an open set $U \subseteq B$. Then a trivialisation $\phi: U \times O(n) \rightarrow p^{-1}(U)$ is given by

$$(x, A) \mapsto \left(\sum_{i=1}^n a_{i1} e_i(x), \dots, \sum_{i=1}^n a_{in} e_i(x) \right),$$

where $A = (a_{ij})$, i.e. $\phi(x, A)$ is the basis of $E_x = p^{-1}(x)$, whose coordinates in the frame (e_i) are precisely the columns of the matrix A . The transition functions of these trivialisations agree with the transition functions of the trivialisations of E . Again, since the transition functions uniquely determine the bundle, we conclude that $O(E)$ is in fact isomorphic to the associated principle bundle as defined at the beginning of this section.

- ii) An orientation of a real vector bundle allows us to further reduce the structure group to $SO(n)$. Similiarly, the associated principal bundle is then given by the *oriented orthonormal frame bundle* $SO(E)$ of E , whose fiber over a point $x \in B$ is the space of all oriented ordered orthonormal bases of E_x .
- iii) An example, which will be of relevance later is the following. Let $P \rightarrow B$ be an S^1 -principal bundle. We can change the fiber of P from S^1 to D^2 to obtain another bundle $E \rightarrow B$, where S^1 acts on D^2 by rotations. Since this action just extends the action of $S^1 = \partial D^2$ on itself we see that $P \cong \partial E$. So in particular the total space of any S^1 -principal bundle is null-bordant.

1.2 Universal Bundles and Classifying Spaces

A natural question is which fiber bundles with specified fiber and structure group can exist over a given base space. Again, this classification problem turns out to be easier for principal bundles. For simplicity we assume all base spaces to be CW-complexes in this section.

Definition 1.7. A G -principal bundle $E \rightarrow B$ is called *universal* if E is weakly contractible as a topological space.

The significance of this definition is established by the following theorem. We say that two morphisms of G -principal bundles are G -homotopic if they are homotopic through morphisms of G -principal bundles.

Theorem 1.8. *Let $E \rightarrow B$ be a universal G -principal bundle. Then for any other G -principal bundle $P \rightarrow X$ there exists a morphism $P \rightarrow E$ of G -principal bundles, which is unique up to G -homotopy.*

Before showing the theorem let us discuss its implications. Firstly we can equivalently formulate the theorem by saying that $E \rightarrow B$ is a terminal object in the homotopy category of G -principal bundles over CW-complexes and thus in particular uniquely determined (up to G -homotopy equivalence). This justifies writing $EG := E$ and $BG := B$ and we refer to $EG \rightarrow BG$ as *the universal G -principal bundle*. The base space BG is called the *classifying space* of G . Since any bundle morphism $(F, f): P \rightarrow EG$ induces a morphism $P \rightarrow f^*EG$ (and vice versa, which is true for arbitrary fiber bundles) we conclude from Proposition 1.3 that

$$P \cong f^*EG.$$

We call f a *classifying map* for P . By homotopy invariance of the pullback we get a well-defined bijection

$$\phi_X: [X, BG] \rightarrow \mathcal{P}_G(X), [f] \mapsto [f^*EG],$$

where $\mathcal{P}_G(X)$ denotes the set of isomorphism classes of G -principal bundles over X . To see that ϕ_X is indeed bijective note that the existence result in Theorem 1.8 shows that ϕ_X is surjective and the uniqueness result shows that it is injective. Moreover ϕ_X is clearly natural in X . Thus we have shown.

Corollary 1.9. *The maps ϕ_X define a natural isomorphism of contravariant functors*

$$[-, BG], \mathcal{P}_G(-): \text{HCW} \rightarrow \text{Set},$$

where HCW denotes the homotopy category of CW-complexes.

Theorem 1.8 will be a simple application of the following theorem.

Theorem 1.10. *Let $P \rightarrow X$ be a G -principal bundle and $A \subseteq X$ a subcomplex. Then any morphism $P|_A \rightarrow EG$ of G -principal bundles can be extended to a morphism $P \rightarrow EG$ of G -principal bundles.*

The proof of this Theorem based on an inductive cell by cell approach. We refer the reader to [?, Theorem 2.7.1].

Proof of Theorem 1.8. Applying Theorem 1.10 to $A = \emptyset$ shows the existence of a morphism $P \rightarrow EG$. Now suppose we are given morphisms $F_0, F_1: P \rightarrow EG$. Consider the G -principal bundle

$$\pi \times \text{id}: P \times [0, 1] \rightarrow X \times [0, 1],$$

where $\pi: P \rightarrow X$ is the bundle projection (this is just the pullback bundle of P under the projection $X \times [0, 1] \rightarrow X$). Then F_0 and F_1 define a morphism of G -principal bundles

$$(F_0, F_1): P \times \{0, 1\} = (P \times [0, 1])|_{X \times \{0, 1\}} \rightarrow EG,$$

which by Theorem 1.10 can be extended to $P \times [0, 1]$, thus yielding a G -homotopy between F_0 and F_1 . $\square$

Of course for this theory to be of any use we need such universal bundles to exist. The following general existence theorem is due to Milnor.

Theorem 1.11 (OG Milnor paper, Ebert). *For any topological group G there exists a universal G -principal bundle $EG \rightarrow BG$, where BG is a CW-complex.*

A priori the space BG constructed in [OG Milnor paper, Ebert] may not be a CW complex. However, if we choose a CW-approximation $f: X \rightarrow BG$ the total space of the bundle $f^*EG \rightarrow X$ is still weakly contractible. This follows by applying the five lemma to the following diagram.

$$\begin{array}{ccccccccc} \pi_{k+1}(X) & \longrightarrow & \pi_k(G) & \longrightarrow & \pi_k(f^*EG) & \longrightarrow & \pi_k(X) & \longrightarrow & \pi_{k-1}(G) \\ \downarrow \cong & & \downarrow \text{id} & & \downarrow & & \downarrow \cong & & \downarrow \text{id} \\ \pi_{k+1}(BG) & \longrightarrow & \pi_k(G) & \longrightarrow & \pi_k(EG) & \longrightarrow & \pi_k(BG) & \longrightarrow & \pi_{k-1}(G) \end{array}$$

We can now use the classification results for principal bundles to obtain analogous results for general fiber bundles. As discussed in section 1.1 we can generally recover information about arbitrary fiber bundles by changing the fiber.

Corollary 1.12. *Let G be a topological group acting effectively on a space F . Then for any CW-complex X the map*

$$[X, BG] \rightarrow \mathcal{F}_{F,G}(X), \quad [f] \mapsto [f^*(EG \times_G F)]$$

is a natural bijection, where $\mathcal{F}_{F,G}(X)$ denotes the set of isomorphism classes of fiber bundles with fiber F and structure group G over X .

Proof. By 1.9 the map

$$[X, BG] \rightarrow \mathcal{P}_G(X), \quad [f] \mapsto [f^*EG]$$

is a natural bijection and by Lemma 1.5

$$\mathcal{P}_G(X) \rightarrow \mathcal{F}_{F,G}(X), \quad [P] \mapsto [P \times_G F]$$

is a natural bijection. Composing both thus yields the natural bijection

$$[X, BG] \rightarrow \mathcal{F}_{F,G}(X), \quad [f] \mapsto [(f^*EG) \times_G F].$$

So the statement follows by again applying Lemma 1.5. $\square$

Again let us discuss a few examples.

Example 1.13.

- (i) The universal cover $\mathbb{R} \rightarrow S^1$ is a $\mathbb{Z}$ -principal bundle, $\mathbb{Z}$ acting on itself by addition. Since $\mathbb{R}$ is contractible we obtain $B\mathbb{Z} \cong S^1$ and $E\mathbb{Z} \cong \mathbb{R}$.
- (ii) From the long exact sequence of homotopy groups

$$\pi_k(EG) \longrightarrow \pi_k(BG) \longrightarrow \pi_{k-1}(G) \longrightarrow \pi_{k-1}(EG)$$

we conclude that $\pi_k(BG) \cong \pi_{k-1}(G)$ for $k \geq 1$. Moreover, BG is connected since it is the image of the connected space EG under $EG \rightarrow BG$.

- (iii) The universal $O(k)$ -principal bundle $EO(k) \rightarrow BO(k)$ also has an explicit description. Consider the Stiefel manifold $V_k^n \subseteq \mathbb{R}^{n \times k}$, which is defined as the compact space of orthonormal k -tuples of vectors in $\mathbb{R}^n$. Moreover let Gr_k^n be the Grassmanian of k -planes in $\mathbb{R}^n$, i.e. the set of k -dimensional linear subspaces of $\mathbb{R}^n$. The topology on Gr_k^n is induced by the projection map

$$p: V_k^n \rightarrow \text{Gr}_k^n, \quad (v_1, \dots, v_k) \mapsto \text{span}(v_1, \dots, v_k)$$

Identifying a linear subspace $V \in \text{Gr}_k^n$ with $\mathbb{R}^k$ yields an identification of the fiber $p^{-1}(V)$ with $O(k)$. Moreover the identifications $V \cong \mathbb{R}^k$ can be chosen consistently in a small neighbourhood around V , providing local trivialisations for p . The transition of two such trivialisations is realized by actions of $O(k)$, hence p defines an $O(k)$ -principal bundle. Now the canonical inclusion $\mathbb{R}^n \rightarrow \mathbb{R}^{n+1}$ induces inclusions $V_k^n \rightarrow V_k^{n+1}$ and $\text{Gr}_k^n \rightarrow \text{Gr}_k^{n+1}$. Consider the colimits over these inclusions

$$V_k^\infty := \text{colim}_n (V_k^n), \quad \text{Gr}_k^\infty := \text{colim}_n (\text{Gr}_k^n)$$

and the induced map $V_k^\infty \rightarrow \text{Gr}_k^\infty$. One can choose trivialisations of $V_k^n \rightarrow \text{Gr}_k^n$ for varying n , which fit together to induce a trivialisation in the colimit, such that $V_k^\infty \rightarrow \text{Gr}_k^\infty$ defines an $O(k)$ -principal bundle. We claim that V_k^∞ is weakly contractible. Consider the map

$$V_k^n \rightarrow S^{n-1}, (v_1, \dots, v_k) \mapsto v_1.$$

Similar arguments as above shows that this map is locally trivial with fiber V_{k-1}^{n-1} . Hence we can consider the long exact sequence of homotopy groups

$$\pi_{i+1}(S^{n-1}) \longrightarrow \pi_i(V_{k-1}^{n-1}) \longrightarrow \pi_i(V_k^n) \longrightarrow \pi_i(S^{n-1})$$

So $\pi_i(V_k^n) \cong \pi_i(V_{k-1}^{n-1})$ for $i \leq n-3$. Applying this inductively yields

$$\pi_i(V_k^n) \cong \pi_i(V_1^{n-(k-1)}) = \pi_i(S^{n-k+1}) = 0$$

for $i \leq n-k-1$. Now from compactness of S^i and the fact that V_k^n is closed in V_k^{n+1} we conclude that the image of any map $S^i \rightarrow V_k^\infty$ is contained in some finite stage V_k^n . Therefore $\pi_i(V_k^\infty) = 0$ for all $i \geq 0$, such that the universal $O(k)$ -principal bundle is explicitly given by $V_k^\infty \rightarrow \text{Gr}_k^\infty$.

Similarly we get an explicit description of the universal $SO(k)$ -principal bundle. Let $\tilde{\text{Gr}}_k^n$ be the set of tuples consisting of a k -dimensional linear subspace of $\mathbb{R}^n$ together with a chosen orientation. The topology on $\tilde{\text{Gr}}_k^n$ is induced by

$$p: V_k^n \rightarrow \tilde{\text{Gr}}_k^n, (v_1, \dots, v_k) \mapsto (\text{span}(v_1, \dots, v_k), \text{o}_{(v_1, \dots, v_k)}),$$

where $\text{o}_{(v_1, \dots, v_k)}$ is the unique orientation on $\text{span}(v_1, \dots, v_k)$, such that $(v_1, \dots, v_k)$ is positively oriented. Similar to above we see that p defines an $SO(k)$ -principal bundle. Taking the colimit, we obtain the universal $SO(k)$ -principal bundle $V_k^\infty \rightarrow \tilde{\text{Gr}}_k^\infty$.

- (iv) Any real k -dimensional vector bundle $E \rightarrow X$ can be considered as a fiber bundle with fiber $\mathbb{R}^k$ and structure group $O(k)$. So Corollary 1.12 and the example above show that every such E is the pullback of the universal real k -dimensional vector bundle $V_k^\infty \times_{O(k)} \mathbb{R}^k$ under some map $X \rightarrow \text{Gr}_k^\infty$. We show that this universal vector bundle also has a more explicit description, namely as the tautological bundle $\gamma_k \rightarrow \text{Gr}_k^\infty$. Recall that this bundle is defined as

$$\gamma_k := \{(X, v) \in \text{Gr}_k^\infty \times \mathbb{R}^\infty \mid v \in X\}$$

together with the canonical projection map to Gr_k^∞ . An element in V_k^∞ is a k -tuple of orthonormal vectors in some $\mathbb{R}^n$, which we will subsequently regard as the columns of an $n \times k$ matrix. Then the map

$$V_k^\infty \times \mathbb{R}^k \rightarrow \gamma_k, (A, v) \mapsto (\text{im}(A), Av)$$

descends to the quotient, defining a vector bundle map $V_k^\infty \times_{O(k)} \mathbb{R}^k \rightarrow \gamma_k$. This map is clearly a linear isomorphism on fibers, hence an isomorphism of vector bundles.

Similarly, oriented real k -dimensional vector bundles are classified by maps to $\tilde{\text{Gr}}_k^\infty$. We claim that the universal oriented real k -dimensional vector bundle is given by the tautological bundle $\gamma_k^{SO} \rightarrow \tilde{\text{Gr}}_k^\infty$ defined by

$$\gamma_k^{SO} := \{(X, \text{o}_X, v) \in \tilde{\text{Gr}}_k^\infty \times \mathbb{R}^k \mid v \in X\}$$

together with the canonical projection map to $\tilde{\text{Gr}}_k^\infty$. To this end consider the map

$$V_k^\infty \times \mathbb{R}^k \rightarrow \gamma_k^{SO}, (A, v) \mapsto (\text{im}(A), \text{o}_A, Av),$$

where o_A is the unique orientation on $\text{im}(A)$, such that $A: \mathbb{R}^k \rightarrow \text{im}(A)$ is orientation preserving. Again this map descends to the quotient defining an isomorphism $V_k^\infty \times_{SO(k)} \mathbb{R}^k \rightarrow \gamma_k^{SO}$.

- (v) Completely analogous to the real case we can also define the complex Stiefel manifolds $V_k^\infty(\mathbb{C})$ and complex Grassmanians $\text{Gr}_k^\infty(\mathbb{C})$ and show that the universal $U(k)$ -principal bundle is given by $V_k^\infty(\mathbb{C}) \rightarrow \text{Gr}_k^\infty(\mathbb{C})$. The universal complex k -dimensional vector bundle is the tautological bundle $\tau_k \rightarrow \text{Gr}_k^\infty(\mathbb{C})$.

- (vi) As special cases of the preceding examples we obtain

$$B\mathbb{Z}_2 \cong \mathbb{R}P^\infty, \quad BS^1 \cong \mathbb{C}P^\infty.$$

1.3 Stiefel-Whitney and Chern-Classes

In the preceeding Chapter we discussed the question of how bundles can be classified. Another fundamental question is of course how to determine, whether two given bundles are isomorphic. As usual in topology, a partial solution to this problem is provided by defining computable invariants. For vector bundles the most important invariants are so called *characteristic classes*. The general definition of a characteristic class is as follows. Let $\mathbb{K} = \mathbb{R}, \mathbb{C}$ and write $\text{Vect}_{\mathbb{K}}(B)$ for the set of isomorphism classes of $\mathbb{K}$ -vector bundles over a space B . Then $\text{Vect}_{\mathbb{K}}(-)$ defines a contravariant functor $\text{Top} \rightarrow \text{Set}$. A characteristic class c for $\mathbb{K}$ -vector bundles with values in the cohomology theory H^* is a natural transformation from the functor $\text{Vect}_{\mathbb{K}}(-)$ to the functor H^* regarded as a functor with values in Set . So in more concrete terms c assigns to each $\mathbb{K}$ -vector bundle $E \rightarrow B$ a cohomology class $c(E) \in H^*(B)$ depending only on the isomorphism class of E , such that for any continuous map $f: B' \rightarrow B$ the following naturality condition holds

$$c(f^*E) = f^*c(E).$$

Therefore, using the theory of classifying spaces developed above, we see that for a real (resp. complex) k -dimensional vector bundles, the characteristic class c is uniquely determined by its value for the universal bundle γ_k (resp. τ_k) discussed in example 1.13 (iv) (resp. ??).

For real vector bundles the most important characteristic classes are the *Stiefel-Whitney classes* and for complex vector bundles the *Chern classes*. Both are formally quite similar and can be characterized by a few axioms.

Definition 1.14 (Stiefel-Whitney Classes, HATCHER). There is a unique sequence of characteristic classes $w_1, w_2, \dots$ for real vector bundles with values in $H^*(-; \mathbb{Z}_2)$, such that

- i) $w_i(E) \in H^i(B; \mathbb{Z}_2)$ for any real bundle $E \rightarrow B$,
- ii) $w_i(E) = 0$ for $i > \text{rank}(E)$,
- iii) $w(E_1 \oplus E_2) = w(E_1) \smile w(E_2)$, where $w = 1 + w_1 + w_2 + \dots$,
- iv) For the canonical line bundle $\gamma_1 \rightarrow \mathbb{R}P^\infty$, $w_1(\gamma_1)$ is the generator of $H^1(\mathbb{R}P^\infty; \mathbb{Z}_2) \cong \mathbb{Z}_2$.

Note that by property ii) the sum $w = 1 + w_1 + w_2 + \dots$ is always finite. The class $w(E)$ is called the *total Stiefel-Whitney class* of E . We sometimes write $w_0(E) := 1 \in H^0(B; \mathbb{Z}_2)$ to simplify notation.

Definition 1.15 (Chern Classes, HATCHER). There is a unique sequence of characteristic classes $c_1, c_2, \dots$ for complex vector bundles with values in $H^*(-; \mathbb{Z})$, such that

- i) $c_i(E) \in H^{2i}(B; \mathbb{Z})$ for any complex bundle $E \rightarrow B$,
- ii) $c_i(E) = 0$ for $i > \text{rank}(E)$,
- iii) $c(E_1 \oplus E_2) = c(E_1) \smile c(E_2)$, where $c = 1 + c_1 + c_2 + \dots$,
- iv) For the canonical line bundle $\tau_1 \rightarrow \mathbb{C}P^\infty$, $c_1(\tau_1)$ is a generator of $H^2(\mathbb{C}P^\infty; \mathbb{Z}) \cong \mathbb{Z}$ specified in advance.

Again the *total Chern class* $c = 1 + c_1 + c_2 + \dots$ is well-defined and we write $c_0(E) := 1 \in H^0(B; \mathbb{Z})$. See [HATCHER] for a proof of existence and uniqueness of the Stiefel-Whitney and Chern classes.

Example 1.16.

- i) Let $\iota: \mathbb{R}P^n \rightarrow \mathbb{R}P^\infty$ be the canonical inclusion. Then the pullback $\iota^*\gamma_1$ is isomorphic to the tautological line bundle $\gamma_1^n \rightarrow \mathbb{R}P^n$. Since ι induces an isomorphism on cohomology we conclude from naturality that $w_1(\gamma_1^n) = \iota^*w_1(\gamma_1)$ is the generator of $H^1(\mathbb{R}P^n; \mathbb{Z}_2) \cong \mathbb{Z}_2$. Similarly $c_1(\tau_1^n)$ of the tautological complex line bundle $\tau_1^n \rightarrow \mathbb{C}P^n$ is a generator of $H^2(\mathbb{C}P^n; \mathbb{Z}) \cong \mathbb{Z}$.
- ii) Let $\varepsilon \rightarrow B$ be a trivial real bundle over some space B . Then $\varepsilon \cong f^*\varepsilon$ for a constant map $f: B \rightarrow B$ and thus $w_i(\varepsilon_n) = f^*w_i(\varepsilon) = 0$ for all $i \geq 1$. Similarly all Chern classes for complex trivial bundles vanish.

- iii) Let $TS^n \rightarrow S^n$ be the tangent bundle of the sphere and let $\nu_n \rightarrow S^n$ be the normal bundle. Since ν_n and $TS^n \oplus \nu_n$ are trivial we conclude from the example above and property iii) in Definition 1.14 that

$$1 = w(TS^n \oplus \nu_n) = w(TS^n) \smile w(\nu_n) = w(TS^n) \smile 1 = w(TS^n).$$

Thus $w_i(TS^n) = 0$ for all $i \geq 1$.

In the last example we have seen that although all Stiefel-Whitney classes of a bundle vanish, the bundle need not be trivial. However for one-dimensional real (resp. complex) bundles it turns out that they are already completely classified by their first Stiefel-Whitney class (resp. Chern class):

Theorem 1.17 (Hatcher, Proposition 3.10). *Let X be a CW-complex. Then the maps*

$$w_1: \text{Vect}_{\mathbb{R}}(X) \rightarrow H^1(X; \mathbb{Z}_2)$$

and

$$c_1: \text{Vect}_{\mathbb{C}}(X) \rightarrow H^2(X; \mathbb{Z})$$

are bijections.

This allows us to define line bundles by only prescribing their first Stiefel-Whitney class (resp. Chern class), which is often technically easier than directly writing down this bundle. While the first Stiefel-Whitney class of higher dimensional bundles does not identify their isomorphism type it does characterize another important invariant, namely orientability:

Proposition 1.18 (Hatcher, Prop 3.11). *Let X be a CW-complex. A real vector bundle $E \rightarrow X$ is orientable if and only if $w_1(E) = 0$.*

Similarly vanishing of the second Stiefel Whitney class $w_2(E)$ corresponds to the existence of another structure on E , namely a spin structure.

Definition 1.19. Let $n \geq 3$.

- (i) The *spin group* $\text{Spin}(n)$ is defined as the universal cover $\rho: \text{Spin}(n) \rightarrow \text{SO}(n)$.
- (ii) Let $E \rightarrow B$ be an oriented real vector bundle equipped with a fiber metric and let $P_{\text{SO}(n)}$ be its oriented orthonormal frame bundle. A *spin structure* on E is a $\text{Spin}(n)$ -principal bundle $P_{\text{Spin}(n)}$ together with a map

$$\phi: P_{\text{Spin}(n)} \rightarrow P_{\text{SO}(n)}$$

of principal bundles, which is ρ -equivariant (see Definition 1.2).

- (iii) An orientable manifold M is called *spin* if its tangent bundle TM admits a spin structure and is called *totally non-spin* if the universal cover of M is not spin.

Note that the existence of a spin structure does not depend on the chosen orientation or fiber metric, since different choices yield isomorphic oriented orthonormal frame bundles. In fact the only obstruction to the existence of a spin structure is the second Stiefel-Whitney class, as mentioned above.

Theorem 1.20 (Lawson, Michelson). *Let X be a CW-complex. An orientable real vector bundle $E \rightarrow X$ admits a spin structure if and only if $w_2(E) = 0$.*

Recall that a manifold M is orientable if and only if its tangent bundle TM restricted to any embedded loop $S^1 \subseteq M$ is trivial. Or in other words “ TM does not contain any Moebius Bands”. A similar characterization can be given for spin manifolds, which is why the property of being spin is sometimes referred to as a higher dimensional analogue of orientability.

Proposition 1.21 (Georg Paper, Proposition 2.6). *Let M be a connected oriented manifold of dimension at least 5.*

- i) *M is spin if and only if for every embedded surface $S \subseteq M$ the restriction $TM|_S$ of the tangent bundle to S is trivial.*

ii) M is totally non-spin if and only if there exists an embedded sphere $S^2 \subseteq M$, such that the restriction $TM|_{S^2}$ of the tangent bundle to this sphere is trivial.

Proof. Since we will not be making use of the first characterization of spin manifolds, we shall only proof part ii) of the proposition and refer the reader to [Georg Paper] for a proof of part i). ... (Proof) $\square$

Remark 1.22. Let $E \rightarrow X$ be a real vector bundle over a CW-complex. If an orientation is given, E can be classified by a map $X \rightarrow BSO(n)$ (see example 1.13). Similiarly, if a spin structure exists, E can be classified by a map $X \rightarrow BSpin(n)$, which can be seen as follows. Let $\phi: P_{Spin(n)} \rightarrow P_{SO(n)}$ be a spin structure on E and let $f: X \rightarrow BSpin(n)$ be the classifying map for $P_{Spin(n)}$. Define a vector bundle

$$\gamma_n^{Spin} := ESpin(n) \times_{Spin(n)} \mathbb{R}^n$$

using Lemma 1.4, where $Spin(n)$ acts on $\mathbb{R}^n$ by virtue of the covering map $\rho: Spin(n) \rightarrow SO(n)$ and the canonical action of $SO(n)$. From Lemma 1.5 we obtain

$$f^* \gamma_n^{Spin} \cong P_{Spin(n)} \times_{Spin(n)} \mathbb{R}^n.$$

Moreover, the spin structure ϕ induces an isomorphism $P_{Spin(n)} \times_{Spin(n)} \mathbb{R}^n \rightarrow P_{SO(n)} \times_{SO(n)} \mathbb{R}^n \cong E$ of vector bundles by

$$[(x, v)] \mapsto [(\phi(x), v)].$$

This map is clearly a linear bijection on fibers, hence an isomorphism of vector bundles.

Finally we record another scenario in which the vanishing of all Stiefel-Whitney classes of a bundle is sufficient for triviality.

Lemma 1.23. *Let $E \rightarrow S$ be a real vector bundle of rank at least 3 over a two dimensional CW-complex S . Then E is trivial if and only if $w_1(E) = w_2(E) = 0$.*

Proof. Combining Theorem 1.20 and Remark 1.22 we conclude that there exists a classifying map $f: S \rightarrow BSpin(n)$ for E , where n is the rank of E . From example 1.13 it follows that $\pi_k(BSpin(n)) = 0$ for $k \geq 2$. Since S is two dimensional the map f is homotopic to a constant ([Hatcher, Lemma 4.6]), therefore $E \cong f^* \xi_{Spin}$ is trivial. $\square$

2 Positive Scalar Curvature

2.1 Overview

brief definition, maybe intuition and obstructions/background etc. based on overview paper

psc is somehow sweet spot for rigidity and flexibility

Probably the most important existence result for positive scalar curvature metrics is the following theorem, which has been shown independently by Schoen and Yau [SY79] as well as Gromov and Lawson [GL80]. We refer the reader to the Appendix for a brief introduction of surgery and cobordism theory.

Theorem 2.1 (Gromov, Lawson, 1980, Chernysh Theorem Georg Paper). *Let M be a compact manifold, which admits a psc metric. Then any manifold M' obtained from M by surgery of codimension at least 3 also admits a psc metric.*

We remark that the construction of the psc metric on M' only changes the given psc metric on M locally near the embedding along which the surgery is performed. In order to apply this theorem we need a way to control the dimension of the surgeries performed or equivalently the indices of handles in a given cobordism. This question is completely independent of scalar curvature and has for example been studied quite thoroughly in the context of the h -cobordism theorem. One central tool to control the indices of the handles in a cobordism is the following theorem due to Wall

Theorem 2.2 (Wall, 1971). (*Wall Theorem*)

Combining the above theorem with the Surgery Theorem 2.1, we can make the following observation. If M_0 admits a psc metric and the inclusion $M_1 \rightarrow W$ is 2-connected, then M_1 also admits a psc metric. In fact, using the conclusion of Theorem 2.2 it is possible to obtain an even stronger result. We say that a metric g on a manifold M with boundary is of *product form near the boundary*, if there is some collar neighbourhood $\iota: \partial M \times [0, 1) \rightarrow M$, such that $\iota^*g = g|_{\partial M} + dt^2$, where $g|_{\partial M}$ is the induced metric on ∂M .

Theorem 2.3 (extending psc to cobordism, Georg, Gaja). *Let W^{n+1} be a cobordism from M_0 to M_1 , such that the inclusion $M_1 \rightarrow W$ is 2-connected. Then any psc metric on M_0 can be extended to a psc metric on W , which is of product form near the boundary.*

2.2 Existence Results for psc metrics

In view of the above discussion, one strategy to construct a psc metric on a given closed manifold M_1 of dimension at least 5 is to first find a cobordism W from some psc manifold M_0 to M_1 and then modify this cobordism by surgeries to obtain another cobordism W' from M_0 to M_1 , such that the inclusion $M_1 \rightarrow W'$ is 2-connected. Of course, such a modification by surgeries may not always be possible, so our next goal is to explain certain scenarios in which it is, one in the spin case and one in the totally non-spin case (see Definition ??).

Theorem 2.4. *Let W^{n+1} be a cobordism from M_0 to M_1 , $n \geq 5$, such that W is spin. Suppose that M_1 is connected, the inclusion $M_1 \rightarrow W$ induces an isomorphism on π_1 and that M_0 carries a psc-metric. Then M_1 also carries a psc-metric. In fact the psc-metric on M_0 can be extended to a psc-metric on W , which is of product form near the boundary.*

Theorem 2.5. *Let W^{n+1} be an oriented cobordism from M_0 to M_1 , $n \geq 5$. Suppose that M_1 is a connected totally non-spin manifold, the inclusion $M_1 \rightarrow W$ induces an isomorphism on π_1 and that M_0 carries a psc-metric. Then M_1 also carries a psc-metric. In fact the psc-metric on M_0 can be extended to a psc-metric on W , which is of product form near the boundary.*

We first show the totally non-spin case.

Proof of Theorem 2.5. Without loss of generality assume W to be connected as well (otherwise only consider the connected component containing M). We first show that $\pi_2(W, M)$ is finitely generated. Let $\tilde{M}$ and $\tilde{W}$ be the universal covers of M and W respectively. Then the inclusion $M \rightarrow W$ lifts to a map $\tilde{M} \rightarrow \tilde{W}$ by virtue of which we can consider the relative homotopy group $\pi_2(\tilde{W}, \tilde{M})$. It follows from the Hurewicz Theorem that

$$\pi_2(\tilde{W}, \tilde{M}) \cong H_2(\tilde{W}, \tilde{M}).$$

By ??? $H_2(\tilde{W}, \tilde{M})$ is finitely generated as a $\mathbb{Z}[\pi_1(M)]$ module (??) hence the same holds for $\pi_2(\tilde{W}, \tilde{M})$. Now consider the following diagram where the rows are exact.

$$\begin{array}{ccccc} \pi_2(\tilde{M}) & \longrightarrow & \pi_2(\tilde{W}) & \twoheadrightarrow & \pi_2(\tilde{W}, \tilde{M}) \\ \downarrow \cong & & \downarrow \cong & & \downarrow \\ \pi_2(M) & \longrightarrow & \pi_2(W) & \twoheadrightarrow & \pi_2(W, M) \end{array}$$

The two rightmost horizontal maps are surjective by exactness, since $\pi_1(\tilde{M}) = 0$ and $\pi_1(M) \rightarrow \pi_1(W)$ is an isomorphism. A diagram chase now shows that $\pi_2(\tilde{W}, \tilde{M}) \rightarrow \pi_2(W, M)$ is an isomorphism.

Next find lifts with... Let

$$F: W \rightarrow B := BSO(n+1)$$

be a classifying map for the tangent bundle TW and set $f := F \circ i$, where $i: M_1 \rightarrow W$ is the inclusion. Then f is a classifying map for $TM_1 \oplus \mathbb{R}$ with $\mathbb{R}$ the trivial line bundle. We claim that f is surjective on π_2 . Since $\pi_2(B) \cong \pi_1(SO(n+1)) \cong \mathbb{Z}_2$ it suffices to show that $\pi_2(f)$ is non-zero. By Proposition 1.21 there exists an embedding $\iota: S^2 \rightarrow M_1$, such that $w_2(\iota^*TM_1) \neq 0$. We compute

$$w_2((f \circ \iota)^*\gamma_n^{\text{SO}}) = w_2(\iota^*(TM_1 \oplus \mathbb{R})) = w_2(\iota^*TM_1) \neq 0.$$

Therefore $f \circ \iota$ represents a non-trivial element in $\pi_2(B)$ showing the claim. Now consider the following diagram, where the row is exact.

$$\begin{array}{ccccc} \pi_2(M_1) & \xrightarrow{i_*} & \pi_2(W) & \longrightarrow & \pi_2(W, M_1) \\ & \searrow f_* & \downarrow F_* & & \\ & & \pi_2(B) & & \end{array}$$

Using surjectivity of f_* , a diagram chase shows that we can find lifts $\alpha_1, \dots, \alpha_l \in \pi_2(W)$ of the generators $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}} \in \pi_2(W, M_1)$, which lie in the kernel of F_* . Moreover, since $n \geq 5$ the classes α_i can be represented by embeddings

$$\phi_i: S^2 \rightarrow W.$$

We show that the normal bundle ν_i of ϕ_i is trivial. Since W is oriented it follows from Proposition 1.18 that $w_1(\phi_i^*TW) = 0$. Furthermore, $F \circ \phi_i$ is null-homotopic by construction, such that

$$w_2(\phi_i^*TW) = w_2((F \circ \phi_i)^*\gamma_n^{\text{SO}}) = 0.$$

Using $w_k(TS^2) = 0$ (see Example 1.16 iii)) we compute

$$w_k(\nu_i) = w_k(\nu_i \oplus TS^2) = w_k(\phi_i^*TW) = 0$$

for $k = 1, 2$. Thus Lemma 1.23 shows that ν_i is indeed trivial. This allows us to perform surgery to obtain a new cobordism W' from M_0 to M_1 with

$$\pi_2(W', M_1) = 0$$

and $\pi_1(W', M_1) \cong \pi_1(W, M_1) = 0$, see Proposition ?? ???. In other words, the inclusion $M_1 \rightarrow W'$ is 2-connected. So by Theorem 2.3 there exists a psc metric g on W' , which is of product form near the boundary. To construct a psc metric on the original cobordism W , observe that since W' was obtained from W by surgeries of dimension 2, W can be obtained from W' by surgeries of codimension 3. Therefore the surgery theorem 2.1 applies to construct a psc metric on W . Since this construction only changes the given metric g locally near the surgery embeddings, the resulting metric on W is still of product form near the boundary. $\square$

Proof of Theorem 2.4. Again $\pi_2(W, M)$ is finitely generated as a $\mathbb{Z}[\pi_1(M)]$ -module. Let $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}}$ be such generators. We could argue as in the proof of Theorem 2.5, replacing $BSO(n+1)$ by $B\text{Spin}(n+1)$ and observing that the corresponding f is trivially surjective on π_2 as $\pi_2(B\text{Spin}(n+1)) \cong \pi_1(\text{Spin}(n+1)) = 0$. However this just amounts to the following more direct argument. Choose lifts $\alpha_1, \dots, \alpha_l \in \pi_2(W)$ of $\alpha_1^{\text{rel}}, \dots, \alpha_l^{\text{rel}}$ and represent α_i by an embedding $\phi_i: S^2 \rightarrow W$. Then

$$w_k(\phi_i^*TW) = \phi_i^*w_k(TW) = 0$$

for $k = 1, 2$ since W is spin. The rest of the proof is now as in Theorem 2.5. $\square$

Remark 2.6. The proofs of Theorem 2.4 and 2.5 are very similar and in fact both statements are a consequence of a more general result. This unifying approach can be formulated using the language of tangential 2-types, which we shall not discuss further in this thesis. See for example [Georg Paper etc.].

In the main construction in Chapter 4 we will invoke Theorem 2.5 to construct a psc metric, so let us conclude this section by discussing some basic results for identifying totally non-spin manifolds.

Lemma 2.7. (i) *The product of a totally non-spin manifold with any non-empty manifold is totally non-spin.*

(ii) *A manifold containing a totally non-spin submanifold of codimension 0 is totally non-spin.*

(iii) *If M is totally non-spin and of dimension $n \geq 3$, then $M - D^n$ is totally non-spin for any embedded disk $D^n \subseteq \text{int}(M)$ in the interior of M .*

Remark 2.8. All statements of the above lemma remain valid if “totally non-spin” is replaced by “non-spin”. The corresponding proofs are just simpler versions of the arguments given below.

Proof. For (i) let M_1 and M_2 be manifolds with M_1 totally non-spin. Then the universal cover of the product $M_1 \times M_2$ is given by the product $\tilde{M}_1 \times \tilde{M}_2$ of the respective universal covers. Denote by

$$\pi_1: \tilde{M}_1 \times \tilde{M}_2 \rightarrow \tilde{M}_1, \quad \pi_2: \tilde{M}_1 \times \tilde{M}_2 \rightarrow \tilde{M}_2$$

the projections and let $\iota: \tilde{M}_1 \rightarrow \tilde{M}_1 \times \tilde{M}_2$ be an inclusion for some choice of basepoint in $\tilde{M}_2$. We compute

$$\begin{aligned} \iota^* w_2(T(\tilde{M}_1 \times \tilde{M}_2)) &= \iota^* w_2(\pi_1^* T\tilde{M}_1 \oplus \pi_2^* T\tilde{M}_2) \\ &= w_2((\pi_1 \circ \iota)^* T\tilde{M}_1 \oplus (\pi_2 \circ \iota)^* T\tilde{M}_2) \\ &= w_2(T\tilde{M}_1) \neq 0, \end{aligned}$$

where we used that $\pi_1 \circ \iota$ is the identity and that $\pi_2 \circ \iota$ is constant, such that $(\pi_2 \circ \iota)^* T\tilde{M}_2$ is trivial. Thus $w_2(T(\tilde{M}_1 \times \tilde{M}_2)) \neq 0$.

For (ii) suppose that $N \subseteq M$ is some totally non-spin codimension 0 submanifold of the manifold M . Let $\iota: N \rightarrow M$ be the inclusion and consider its lift to the universal covers:

$$\begin{array}{ccc} \tilde{N} & \xrightarrow{\tilde{\iota}} & \tilde{M} \\ \downarrow & & \downarrow \\ N & \xrightarrow{\iota} & M \end{array}$$

Since $\tilde{\iota}$ is a local diffeomorphism, the differential $d\tilde{\iota}: T\tilde{N} \rightarrow T\tilde{M}$ is a fiber-wise isomorphism, such that the following is a pullback diagram

$$\begin{array}{ccc} T\tilde{N} & \xrightarrow{d\tilde{\iota}} & T\tilde{M} \\ \downarrow & & \downarrow \\ \tilde{N} & \xrightarrow{\tilde{\iota}} & \tilde{M} \end{array}$$

Therefore $\tilde{\iota}^* w_2(T\tilde{M}) = w_2(T\tilde{N}) \neq 0$ and so $w_2(T\tilde{M}) \neq 0$.

For (iii) assume without loss generality that M is connected. Denote by $B^n := \text{int}(D^n)$ the open ball. Then $M - D^n$ is the interior of $M - B^n$ from which we conclude that $M - D^n$ is totally non-spin if and only if $M - B^n$ is totally non-spin. This follows using an analogous argument as in (ii) by noting that for a manifold X with boundary the universal cover of $\text{int}(X)$ is given by $\text{int}(\tilde{X})$ and the inclusion $\text{int}(\tilde{X}) \rightarrow \tilde{X}$ is a homotopy equivalence. We will now show that $M - B^n$ is totally non-spin. Let $p: \tilde{M} \rightarrow M$ be the universal cover of M and write $M_0 := M - B^n$. We claim that the universal cover of M_0 agrees with the covering

$$\hat{M}_0 = \tilde{M}|_{M_0} = \tilde{M} - p^{-1}(B^n) \xrightarrow{p} M_0.$$

First of all $\hat{M}_0$ is connected as $n \geq 2$. We now have to show that $\hat{M}_0$ is simply connected. Let $F = p^{-1}(x)$ be some fiber of p . From the classification of coverings we see that

$$p^{-1}(B^n) \cong B^n \times F$$

since B^n is simply connected. If the fiber F is finite we can apply the Seifert-van-Kampen Theorem to conclude that $\tilde{M}$ with a single disk B^n removed is still simply connected, such that the statement follows inductively. If the fiber is not finite we can argue more generally as follows. Let $\phi: S^1 \rightarrow \hat{M}_0$ be some smooth loop. Certainly we can contract this loop in $\tilde{M}$, that is we can find some map $\Phi: D^2 \rightarrow \tilde{M}$ restricting to α on the boundary. We now show that Φ is homotopic rel S^1 to some map $\Phi_1: D^2 \rightarrow \tilde{M}$, such that the image of Φ_1 is already contained in $\hat{M}_0$, from which the claim follows. To construct such a homotopy let $0 \in B^n \subseteq M$ be the center of the embedded disk. After a preliminary homotopy we may assume that Φ is transversal to $p^{-1}(0)$. The crucial observation is that, since $n \geq 3$, this is equivalent to requiring that Φ does not intersect $p^{-1}(0)$. Now note that S^{n-1} is a strong deformation retract of $D^n - 0$ and applying such a deformation retraction on each disk $D^n - 0 \subseteq p^{-1}(D^n - 0)$ shows that $\hat{M}_0$ is a strong deformation retract of $\tilde{M} - p^{-1}(0)$. Let

$$r: \tilde{M} - p^{-1}(0) \rightarrow \hat{M}_0$$

be such a retraction and let

$$r_t: \tilde{M} - p^{-1}(0) \rightarrow \tilde{M} - p^{-1}(0)$$

be a homotopy rel S^1 from the identity to r (composed with the corresponding inclusion). Then $\Phi_t := r_t \circ \Phi$ defines a homotopy rel S^1 from Φ to some map Φ_1 whose image is contained in $\hat{M}_0$. This shows the claim and hence the universal cover of M_0 is given by

$$\hat{M}_0 = \tilde{M} - p^{-1}(B^n) \cong \tilde{M} - (B^n \times F).$$

Finally, the Mayer-Vietoris sequence of the pushout

$$\begin{array}{ccc} S^{n-1} \times F & \longrightarrow & D^n \times F \\ \downarrow & & \downarrow \\ \tilde{M} - (B^n \times F) & \xrightarrow{\iota} & \tilde{M} \end{array}$$

where all maps are inclusions, then shows that ι is injective on $H^2(-; \mathbb{Z}_2)$. Therefore

$$w_2(T\hat{M}_0) = w_2(\iota^*T\tilde{M}) = \iota^*w_2(T\tilde{M}) \neq 0.$$

□

A standard example of a non-spin manifold is $\mathbb{C}P^2$. Since $\mathbb{C}P^2$ is simply connected this is also an example of a totally non-spin manifold, which we will appeal to in section ...

Lemma 2.9. $\mathbb{C}P^2$ is non-spin.

(proof)

3 Enlargeability

In this section we discuss the concept of enlargeability, which is the central idea the results in this thesis are concerned with.

Definition 3.1. A closed oriented Riemannian n -manifold M is called enlargeable if for all $\varepsilon > 0$ there exists a Riemannian covering $\tilde{M} \rightarrow M$ together with an ε -contracting (that is ε -Lipschitz) map $\tilde{M} \rightarrow S^n$, which is constant outside a compact subset and of non-zero degree.

Note that this notion does not depend on the particular choice of Riemannian metric on M , since for closed manifolds all metrics are in bi-Lipschitz correspondence. Intuitively, the covering $\tilde{M}$ can be pictured as an “expansion” of M itself (think for example of the usual sketches of coverings of S^1). Enlargeability now means that this expansion can be chosen arbitrarily large, such that it can be wrapped as tightly around a sphere as one wishes. Naively one might expect that for an enlargeable manifold M the universal cover $\tilde{M}$ - which in some sense is the “largest” covering - is hyperspherical, meaning that it admits an ε -contracting map $\tilde{M} \rightarrow S^n$, which is constant outside a compact subset and of non-zero degree for arbitrarily small $\varepsilon > 0$. However, Brunnbauer and Hanke have constructed counterexamples to this naive expectation in [BH, Theorem 1.4].

The simplest example of an enlargeable manifold is the Torus T^n , since it is covered by $\mathbb{R}^n$, which is clearly hyperspherical. More generally the same holds for any closed manifold admitting a metric of non-positive sectional curvature by the Cartan-Hadamard Theorem. Non-enlargeable manifolds are for example all simply connected manifolds.

The central fact about enlargeable manifolds is that they do not admit psc metrics.

Theorem 3.2 (GL1980, GL1982, CS2020). *Enlargeable manifolds do not admit metrics of positive scalar curvature.*

This is the result that motivated the original definition by Gromov and Lawson [GL1980], where they first showed that enlargeability is an obstruction to the existence of psc metrics, under the additional assumption that the coverings $\hat{M}$ in Definition 3.1 are finite and spin. They were later able to drop the finiteness assumption [GL1982]. Both proofs are based on techniques using the spin Dirac operator, so it was unclear how and if these results carry over to the non-spin case. Rather recently however, Cecchini and Schick resolved this issue in [CS2020] by relying on modern minimal surface methods developed by Schoen and Yau [SY...].

The following Lemma now provides two easy ways to construct new enlargeable manifolds out of given ones.

Lemma 3.3. (i) *The product of enlargeable manifolds is enlargeable.*

(ii) *A closed oriented manifold admitting a non-zero degree map onto an enlargeable manifold is enlargeable.*

Proof. For ?? let M^m and N^n be enlargeable. For $\varepsilon > 0$ choose Riemannian coverings $p: \hat{M} \rightarrow M$ and $q: \hat{N} \rightarrow N$ as well as ε -contracting maps $f: \hat{M} \rightarrow S^m$ and $g: \hat{N} \rightarrow S^n$ of non-zero degree. Then

$$p \times q: \hat{M} \times \hat{N} \rightarrow M \times N$$

is a Riemannian covering, where both manifolds are equipped with the corresponding product metric. Moreover

$$f \times g: \hat{M} \times \hat{N} \rightarrow S^m \times S^n$$

defines an ε -contracting map of degree $\deg(f \times g) = \deg(f)\deg(g) \neq 0$. Now consider the map

$$\phi: S^m \times S^n \rightarrow S^m \wedge S^n \cong S^{m+n},$$

given by projection onto the smash product

$$S^m \wedge S^n := S^m \times S^n / (S^m \times * \cup * \times S^n),$$

which is well known to be homeomorphic to the sphere S^{m+n} . The map ϕ is easily seen to be of degree ± 1 (depending on the chosen orientations) and after a slight perturbation we may also assume it to be smooth. Moreover, because $S^m \times S^n$ is closed, it follows that ϕ is a λ -Lipschitz

map for some constant $\lambda > 0$. Therefore $\phi \circ (f \times g)$ is a $\lambda\varepsilon$ -Lipschitz map of non-zero degree and since ε was chosen arbitrarily the statement follows.

For ?? let M^m be enlargeable and let $h: P \rightarrow M$ be some non-zero degree map. Again given $\varepsilon > 0$ choose a Riemannian covering $p: \hat{M} \rightarrow M$ as well as an ε -contracting map $f: \hat{M} \rightarrow S^m$. Define a covering $\pi: \hat{P} \rightarrow P$ as the pullback covering of p under h , that is π fits into a pullback diagram of the form

$$\begin{array}{ccc} \hat{P} & \xrightarrow{H} & \hat{M} \\ \downarrow \pi & & \downarrow p \\ P & \xrightarrow{h} & M \end{array}$$

where H is some suitable smooth map. Since $\hat{M}$ and P are closed, the same holds for $\hat{P}$. Moreover h is μ -Lipschitz for some constant $\mu > 0$, because P is closed. Consequently H is also μ -Lipschitz, given that both π and p are local isometries. For the degree of H we compute

$$\deg(p)\deg(H) = \deg(p \circ H) = \deg(h \circ \pi) = \deg(h)\deg(\pi) \neq 0,$$

using the fact that $\deg(h) \neq 0$ by assumption and that the degree of a covering is given by the cardinality of the fiber. We conclude that H is of non-zero degree, such that $f \circ H: \hat{P} \rightarrow S^m$ is a $\mu\varepsilon$ -Lipschitz map of non-zero degree. Again since ε was chosen arbitrarily the statement follows. $\square$

From Part ?? of the above lemma it follows immediately, that enlargeability is a homotopy invariant, since a homotopy equivalence of connected closed oriented manifolds has degree ± 1 . In fact much more is true, namely that enlargeability is a homological invariant in the following sense.

Theorem ([?]). *Let M be a closed oriented n -manifold and let $\phi: M \rightarrow B\pi_1(M)$ be the classifying map of the universal cover. Then M is enlargeable if and only if $\phi_*[M] \in H_n(B\pi_1(M); \mathbb{Q})$ does not lie in the rational vector subspace*

$$H_n^{\text{small}}(B\pi_1(M); \mathbb{Q}) \subseteq H_n(B\pi_1(M); \mathbb{Q})$$

of “small” classes, i.e. $\phi_*[M]$ is “large”.

In [?] the authors showed this by giving a general definition of “small” and “large” homology classes of a simplicial complex X with finitely generated fundamental group, such that the assignment $X \mapsto H_n^{\text{small}}(X; \mathbb{Q})$ is functorial for continuous maps $f: X \rightarrow Y$ inducing an epimorphism on π_1 , that is $f_*: H_n(X; \mathbb{Q}) \rightarrow H_n(Y; \mathbb{Q})$ maps small classes to small classes. Moreover if f happen to induce an isomorphism on π_1 then f_* also maps large classes to large classes. This general definition then specializes to the fact that a closed oriented manifold M is enlargeable if and only if the fundamental class $[M]$ is large, that is $[M] \notin H_n^{\text{small}}(M; \mathbb{Q})$. Combining these observations, the theorem follows. Furthermore this argument shows that the notion of enlargeability can be purely thought of as a concept of simplicial complexes. For details we refer the reader to [?].

4 Construction of the Circle Bundles

We will now construct counterexamples to the question raised in the beginning, that is circle bundles $E \rightarrow M$ over enlargeable M , where the total space E admits a metric of positive scalar curvature.

For $n \geq 2$ let N^{2n} be any $2n$ -dimensional totally non-spin manifold. We could for example take $N := \mathbb{C}P^2 \times T^{2n-4}$, which is totally non-spin by Lemma 2.7 and Lemma 2.9. Now consider the manifold

$$M := \mathbb{C}P^n \# T^{2n} \# N.$$

Since the collapse map $M \rightarrow T^{2n}$ has degree 1 and T^{2n} is enlargeable, it follows from Lemma 3.3 that M is also enlargeable. We will now construct a circle bundle $E \rightarrow M$ as follows. Let

$$c_1 \in H^2(M; \mathbb{Z}) \cong H^2(\mathbb{C}P^n; \mathbb{Z}) \oplus H^2(T^{2n}; \mathbb{Z}) \oplus H^2(N; \mathbb{Z})$$

be the cohomology class corresponding to the first chern class $\alpha \in H^2(\mathbb{C}P^n; \mathbb{Z})$ of the canonical bundle $S^{2n+1} \rightarrow \mathbb{C}P^n$ under the above isomorphism. We then define $E \rightarrow M$ as the unique S^1 -principal bundle with first chern class equal to c_1 . Now this is quite an abstract definition, so let us develop an intuition for how this bundle looks like. We claim that E is trivial over the summand $T^{2n} \# N$ and is given by the restriction of the canonical bundle $S^{2n+1} \rightarrow \mathbb{C}P^n$ to $\mathbb{C}P^n - D^{2n} \subseteq M$ over the $\mathbb{C}P^n$ summand. To make this precise let

$$i_1: (T^{2n} \# N) - D^{2n} \rightarrow M$$

be the inclusion. Then

$$\begin{array}{ccc} H^2(M; \mathbb{Z}) & \xrightarrow{i_1^*} & H^2((T^{2n} \# N) - D^{2n}; \mathbb{Z}) \\ \cong \uparrow & & \cong \uparrow \\ H^2(\mathbb{C}P^n; \mathbb{Z}) \oplus H^2(T^{2n} \# N; \mathbb{Z}) & \xrightarrow{\text{proj.}} & H^2(T^{2n} \# N; \mathbb{Z}) \end{array}$$

commutes and therefore $i_1^*(c_1) = 0$. Again using the fact that S^1 -principal bundles are uniquely determined by their first chern class we see that i_1^*E is indeed trivial. Similarly if we denote by

$$i_2: \mathbb{C}P^n - D^{2n} \rightarrow M, \quad j: \mathbb{C}P^n - D^{2n} \rightarrow \mathbb{C}P^n$$

the inclusions, the diagram

$$\begin{array}{ccc} H^2(M; \mathbb{Z}) & \xrightarrow{i_2^*} & H^2(\mathbb{C}P^n - D^{2n}; \mathbb{Z}) \\ \cong \uparrow & & \cong \uparrow j^* \\ H^2(\mathbb{C}P^n; \mathbb{Z}) \oplus H^2(T^{2n} \# N; \mathbb{Z}) & \xrightarrow{\text{proj.}} & H^2(\mathbb{C}P^n; \mathbb{Z}) \end{array}$$

commutes and thus $i_2^*(c_1) = j^*(\alpha)$. So the first chern classes of i_2^*E and j^*S^{2n+1} agree, hence they must be isomorphic bundles.

With this intuition in mind we will now show the following two claims.

Claim 1. E is totally non-spin.

Claim 2. The projection $p: E \rightarrow M$ induces an isomorphism on π_1 .

Granting these claims for now, let us see how to conclude the existence of a positive scalar curvature metric on E . The disk bundle $DE \rightarrow M$ corresponding to E defines an oriented null-cobordism of E (see Example 1.6 iii)). Moreover, since $DE \rightarrow M$ is a homotopy equivalence it follows from Claim 1 that the inclusion $E \rightarrow DE$ is an isomorphism on π_1 . So together with Claim 2 Theorem 2.5 applies, hence E admits a psc metric.

Let us now proof Claim 1 and Claim 2.

Proof of Claim 1. Let $i: N - D^{2n} \rightarrow M$ be the inclusion. We have already established above, that E is trivial over the $T^{2n} \# N$ summand and so in particular

$$i^*E \cong (N - D^{2n}) \times S^1.$$

From Lemma 2.7 (i) and (iii) we conclude that $(N - D^{2n}) \times S^1$ is totally non-spin. Therefore E contains a totally non-spin submanifold of codimension 0 and so Lemma 2.7 (ii) implies that E is also totally non-spin. $\square$

Proof of Claim 2. We have to show that the fibers of E can be contracted. This is essentially due to the fact that E contains (the restriction of) the canonical bundle $S^{2n+1} \rightarrow \mathbb{C}P^n$ for which the fibers are contractible. To formalize this let $j: \mathbb{C}P^n - D^{2n} \rightarrow \mathbb{C}P^n$ be the inclusion. Consider the following commutative diagram, where the rows are taken from the long exact sequences of homotopy groups of the respective fibrations and the vertical maps are induced by inclusions:

$$\begin{array}{ccccc}
\pi_2(M) & \xrightarrow{\partial_1} & \pi_1(S^1) & \longrightarrow & \pi_1(E) \\
\uparrow & & \uparrow \text{id} & & \uparrow \\
\pi_2(\mathbb{C}P^n - D^{2n}) & \xrightarrow{\partial_2} & \pi_1(S^1) & \longrightarrow & \pi_1(j^*E) \\
\cong \downarrow & & \downarrow \text{id} & & \downarrow \\
\pi_2(\mathbb{C}P^n) & \xrightarrow{\partial_3} & \pi_1(S^1) & \longrightarrow & \pi_1(S^{2n+1})
\end{array}$$

The vertical map on the lower left is an isomorphism, because both $\mathbb{C}P^n$ and $\mathbb{C}P^n - D^{2n}$ are simply connected and so by the Hurewicz Theorem

$$\begin{array}{ccc}
\pi_2(\mathbb{C}P^n - D^{2n}) & \longrightarrow & \pi_2(\mathbb{C}P^n) \\
\downarrow \cong & & \downarrow \cong \\
H_2(\mathbb{C}P^n - D^{2n}; \mathbb{Z}) & \longrightarrow & H_2(\mathbb{C}P^n; \mathbb{Z})
\end{array}$$

commutes and the vertical maps are isomorphisms. Since the bottom horizontal map is an isomorphism the same holds for the top horizontal map. We now claim that ∂_1 is surjective. For this it suffices to show surjectivity of ∂_2 , which in turn is equivalent to surjectivity of ∂_3 . But this is clear by exactness of the rows and $\pi_1(S^{2n+1}) = 0$. Consequently

$$\pi_1(S^1) \rightarrow \pi_1(E)$$

is the zero map and therefore we conclude from exactness of

$$\pi_1(S^1) \rightarrow \pi_1(E) \rightarrow \pi_1(M) \rightarrow \pi_0(S^1),$$

that $\pi_1(E) \rightarrow \pi_1(M)$ is indeed an isomorphism. $\square$

The above construction now gives examples for circle bundles with positive scalar curvature over enlargeable manifolds M in all even dimensions $\dim(M) \geq 4$. To also get examples in odd dimensions let $E \rightarrow M$ be as above and let $\pi: M \times S^1 \rightarrow M$ be the projection. Then the pullback bundle $\pi^*E \cong E \times S^1$ clearly admits a metric of positive scalar curvature and by Lemma 3.3 the base space $M \times S^1$ is still enlargeable. Naturally, the question arises as to what happens in dimensions $\dim(M) \leq 3$. In this case it turns out that no circle bundle $E \rightarrow M$ admits a metric of positive scalar curvature, see [?, Theorem B].